Secondary Examination, 2015

Mathematics

Time- $3\frac{1}{4}$ **Hours**

M.M- 80

GENERAL INSTRUCTIONS TO THE EXAMINEES:

1) Candidate must write first his/ her Roll No. on the question paper compulsorily.
2) All the questions are compulsory.
3) Write the answer to each question in the given answer- book only.
4) For questions having more than one part the answers to those parts are to be written together in continuity.
5) If there is any error/difference/contradiction in Hindi and English versions of the quation paper of Hindi version be treated valid.
6)

Section	Q.Nos.	Marks per question
A	1-10	1
B	11-15	2
C	16-25	3
D	26-30	6

7) There are internal choices in Q. Nos. 28 and 30.

8) Write on both sides of the pages of your answer-book. If any rough work is to be done, do It on last pages of the answer-book and cross with slant lines and write 'Rough Work' on them.

9) Draw the graph of Q. No. 26 on the graph paper.

PART-A

1. Write the common difference of the A.P. $7,\ 5,\ 3,\ 1,\ -1,\ -3,\ .....$

A.P. $7,\ 5,\ 3,\ 1,\ -1,\ -3,\ .....$

Solution:

Given that,

A.P. $7,\ 5,\ 3,\ 1,\ -1,\ -3,\ .....$

$a_2 = 5$

$a_1 = 7$

$d = a_2 - a_1$

$= 5 - 7$

$= -2$

Hence, the common difference is -2.

2. Write the distance of the point (– 5, 4) from x-axis.

Solution:

Consider the distance of the point is (– 5, 4).

Then,

The x-axis is 4.

3. Write the Solution of the pair of linear equations $4x + 2y = 5$ **and** $x - 2y = 0$**.**

Solution:

Consider the given linear equations are

$4x + 2y = 5$-------(1)

$x - 2y = 0$----------(2)

Now,

Adding equation (1) and (2)

$4x + 2y = 5$

$x - 2y = 0$

$$\boxed{5x = 5}$$

$$x = \frac{5}{5}$$

$$x = 1$$

Putting the value of x in equation (i)

$4x + 2y = 5$

$4(1)+2y = 5$

$4 + 2y = 5$

$2y = 5 - 4$

$2y = 1$

$$y = \frac{1}{2}$$

4. Find the HCF of 96 and 404 by the Prime Factorisation Method.

Solution:

Given that,

HCF of 96 and 404

Now, Prime Factorisation Method

$96 = 2^5 \times 3$

$404 = 2^2 \times 101$

$HCL = 2^2 = 4$

5. One card is drawn from a well-shuffled deck of 52 cards. Calculate the probability that the card will not be an ace.

Solution:

Given that,

Total number of cards = 52

Total number of aces = 4

$$P(\text{getting an ace}) = \frac{\text{No. of aces}}{\text{total no. of cards}} = \frac{4}{52} = \frac{1}{13}$$

Then,

$P(\text{not get an ace}) = 1 - P(\text{get an ace})$

$= 1 - \frac{1}{13}$

$= \frac{13-1}{13}$

$= \frac{12}{13}$

6. If K (5, 4) is the mid-point of the line segment PQ and co-ordinates of Q are (2, 3), then find the co-ordinates of point P.

Solution:

Given that,

Q = (2,3)

And

Mid point K=(5,4)

Mid point of coordinate

$= \left(\frac{x_2 + x_1}{2}, \frac{y_2 + y_1}{2} \right)$

$x = \frac{x_2 + x_1}{2}$

$5 = \frac{2 + x_1}{2}$

$10 = 2 + x_1$

$x_1 = 8$

$y = \frac{y_2 + y_1}{2}$

$4 = \frac{3 + y_1}{2}$

$8 = 3 + y_1$

$y_1 = 5$

Hence, the co-ordinates of point P = (8,5)

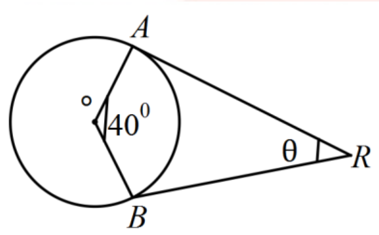
7. If tangents RA and RB from a point R to a circle with center O are inclined to each other at an angle of θ and ∠ AOB = 40° then find the value of θ .

Solution:

Consider,

Angle between Radius and tangent

$A = 90^0$
$B = 90^0$



from $\square OBRA$,

$$\angle AOB + \angle ARB = 180$$
$$40 + \angle ARB = 180$$
$$\angle ARB = 180 - 40$$
$$\angle ARB = 140$$

8. How many tangents can be constructed to any point on the circle of radius 4 cm ?

Solution:

One point = one tangent

9. Find the circumference of a circle whose diameter is 14 cm.

Solution:

Given that,

Diameter = 14 c.m

Then,

Circumference of circle

$$= 2\pi r$$
$$= 2\times\frac{22}{7}\times 14$$
$$= 88cm$$

Hence, the circumference of a circle is 88 cm.

10. Write the length of an arc of a sector of circle with radius r and angle with degree measure θ.

Solution:

Given that,

Radius = r

Angle = θ

Then,

Length of arc $= \frac{\pi r\theta}{180}$

PART-B

11. Show that $\sin 28^{\circ}\cos 62^{\circ} + \cos 28^{\circ}\sin 62^{\circ} = 1.$

Solution:

Consider,

$$\sin 28^{\circ}\cos 62^{\circ} + \cos 28^{\circ}\sin 62^{\circ} = 1.$$
$$\sin\theta = \cos(90-\theta)$$
$$\cos\theta = \sin(90-\theta)$$

Then,

$$\Rightarrow \cos(90-28^{0})\cos 62^{0} + \sin(90-28^{0})$$

12. Find the value of $\frac{\tan 67^0}{\cot 23^0}$**.**

Solution:

Consider

$$\frac{\tan 67^0}{\cot 23^0}$$

$$\because \tan\theta = \cot(90-\theta)$$

$$\therefore \tan\theta = \frac{\cot(90-67^0)}{\cot 23^0}$$

$$= \frac{\cot 23^0}{\cot 23^0}$$

$$= 1$$

13. If $3\cot A = 4$, **then evaluate** $\frac{1-\tan^2 A}{1+\tan^2 A}$.

Solution:

Given that,

$$3\cot A = 4$$

$$\cot A = \frac{4}{3} \qquad \left(\because \cot A = \frac{B}{p}\right)$$

$$\tan A = \frac{3}{4} \qquad \left(\because \tan A = \frac{P}{B}\right)$$

Now evaluate,

$$\frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$= \frac{1-\left(\frac{3}{4}\right)^2}{1+\left(\frac{3}{4}\right)^2}$$

$$= \frac{1-\frac{9}{16}}{1+\frac{9}{16}}$$

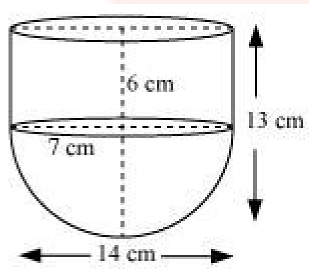
$$= \frac{\frac{16-9}{16}}{\frac{16+9}{16}}$$

$$= \frac{\frac{7}{16}}{\frac{25}{16}} = \frac{7}{25}$$

14. A vessel is in the form of a hollow hemisphere mounted by a hollow cylinder. The radius of the hemisphere is 7 cm and the total height of the vessel is 13 cm. Find the inner surface area of the vessel.

Solution:

Consider



Height of Hemispherical part (r) = 7 cm

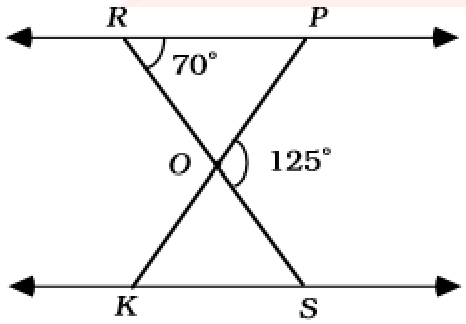
Height of cylindrical part (h) = 13-7 = 6 cm

Inner surface area of the vessel

= CSA of cylindrical part + CSA of hemispherical part

$= 2\pi rh + 2\pi r^2$
$= 2\pi r(h + r)$
$= 2 \times \frac{22}{7} \times 7(6 + 7)$
$= 44 \times 13$
$= 572 cm^2$

15. In the figure, Δ OPR ~ Δ OSK , $\angle$POS = 125° **and** $\angle$PRO = 70°**. Find the values of** $\angle$OKS **and** $\angle$ROP**.**



Solution:

Given that,

Δ OPR ~ Δ OSK
$\angle$POS = 125°
and,
$\angle$PRO = 70°

Now, the linear pair

$y + 125 = 180$
$y = 180 - 125$
$y = 55$

So, the angle sum properties is

$\angle P + 70 + y = 180$
$\angle P = 180 - 125$
$\angle P = 55$

Therefore,

$\angle$P = $\angle$OKS $\quad (\because \Delta OPR \sim \Delta OBK)$,

$\therefore \angle OKS = 55$

PART – C

16. Prove that $\left[\dfrac{1 - \tan A}{1 - \cot A}\right]^2 = \tan^2 A$

Solution:

Consider

$$L.H.S = \left[\frac{1 - \tan A}{1 - \cot A}\right]^2$$

$$= \left[\frac{1 - \frac{\sin A}{\cos A}}{1 - \frac{\cos A}{\sin A}}\right]^2$$

$$= \frac{\left(\frac{\cos A - \sin A}{\cos A}\right)^2}{\left(\frac{\sin A + \cos A}{\sin A}\right)^2}$$

$$= \frac{(\cos A - \sin A)^2}{\cos^2 A} \times \frac{\sin^2 A}{(\sin A + \cos A)^2}$$

$$= \frac{\sin^2 A}{\cos^2 A}$$

$= \tan^2 A$ RHS Proved.

17. Divide $3x^3 + x^2 + 2x + 5$ **by** $1 + 2x + x^2$ **.**

Solution:

Consider

$$\begin{array}{r|l} & 3x-5 \\ x^2+2x+1 & 3x^3 + x^2 + 2x + 5 \\ & 3x^3 + 6x^2 + 3x \\ & - \quad - \quad - \\ \hline & -5x^2 - x + 5 \\ & -5x^2 - 10x - 5 \\ & + \quad + \quad + \\ \hline & 9x+10 \end{array}$$

18. Prove that $\sqrt{2}$ **is an irrational number.**

Solution:

Let $\sqrt{2}$ is a Rational no.

$\sqrt{2} = \frac{a}{b}$ then, $b \neq 0$

Now, squaring both sides

$$\left(\sqrt{2}\right)^2 = \left(\frac{a}{b}\right)^2$$

$$\frac{2}{1} = \frac{a^2}{b^2}$$

$$2b^2 = a^2 ----(i)$$

$\because a^2$, div by 2

$\therefore a$ also divide by 2 then,

$$\frac{a}{2} = \frac{c}{1}$$

$$a = 2c$$

Put the value of a in equation (i)

$$2b^2 = (2c)^2$$
$$2b^2 = 4c^2$$
$$b^2 = 2c^2$$
$$c^2 = \frac{b^2}{2}$$

$\because b^2$ divisional by 2

$\therefore b$ also divisional by 2

So, $\sqrt{2}$ is a irrational.

19. How many terms of the A.P. 17, 15, 13, ... must be taken, so that their sum is 81 ?

Solution:

Given that,

A.P. 17, 15, 13, ...

$$a = 17$$
$$d = a_2 - a_1$$
$$= 15 - 17$$
$$= -2$$
$$s_n = 81$$

Now, the formula

$$s_n = \frac{n}{2}\left[2a + (n-1)d\right]$$
$$81 = \frac{n}{2}\left[2(17) + (n-1)(-2)\right]$$
$$81 \times 2 = n\left[34 - 2n + 2\right]$$
$$162 = n\left[36 - 2n\right]$$
$$162 = 36n - 2n^2$$
$$2n^2 - 36n + 162 = 0$$

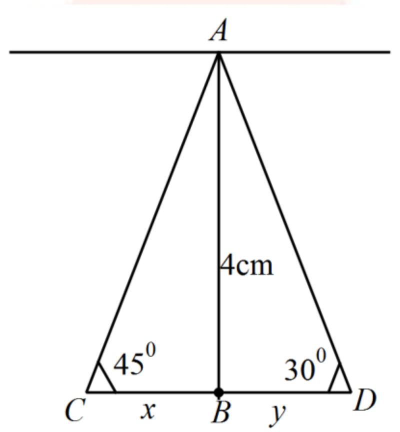
Now, divide 2 on both sides,

$$n^2 - 18n + 81 = 0$$
$$n^2 - 2(n)(9) + (9)^2 = 0$$
$$(n-9)^2 = 0$$
$$(n-2)(n-2) = 0$$
$$n - 2 = 0$$
$$n = 2$$

20. From a point on a bridge across a river the angles of depression of the banks on opposite sides of the river are 30° and 45° respectively. If the bridge is at a height of 4 m from the banks, find the width of the river.

Solution:

Consider



In the triangle ABC

$$\tan 45^0 = \frac{P}{B}$$
$$= 1 = \frac{4}{x}$$
$$x = 4$$

Again, in the triangle ABD

$$\tan 30^0 = \frac{P}{B}$$

$$= \frac{1}{\sqrt{3}} = \frac{4}{y}$$

$$y = \frac{4}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

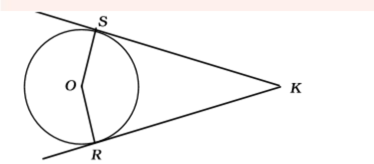
$$y = \frac{4\sqrt{3}}{3}$$

The width of the river,

$$= 4 + \frac{4\sqrt{3}}{3}$$

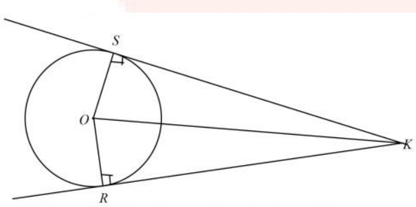
$$= \frac{12 + 4\sqrt{3}}{3} m$$

21. In the given figure, O is the center of a circle and two tangents KR, KS are drawn on the circle from a point K lying outside the circle. Prove that KR = KS.



Solution:

Given that,



KR and KS are tangent drawn to a circle with centre O from an external point K.

Prove that,

KR = KS

Now,

ΔKOS and ΔKOR

$KO = KO$ (By common factor)

SO = RO (By radius of the circle)

$\angle 1 = \angle 2$ (Right angle(90))

RHS,

$\Delta KOS \cong \Delta KOR$

KR = KS (By CPCT)

22. Construct a triangle of sides 4 cm, 5 cm and 6 cm and then a triangle similar to it whose sides are $\frac{3}{5}$ time of the corresponding sides of the

Solution:

Given that,

Triangle of sides

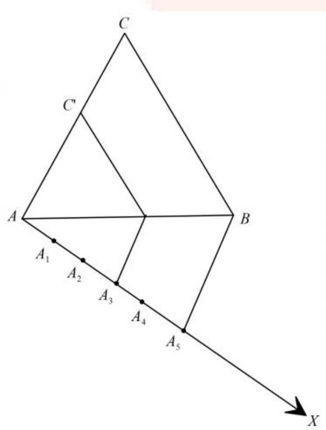
4cm, 5cm, 6cm

And then a triangle similar to whose sides are

$\left(\frac{3}{5}\right)$

Of the corresponding sides of it.

We follow the following steps



Step-1 First of all we draw a line segment

$AB = 4cm$

Step-2 With A as centre and radius

$AC = 6cm$

Step-3 With B as centre and radius

$BC = 5cm$

Draw an arc, intersecting the arc drawn in step-2 at C.

Step-4 Join AC and BC to obtain

ΔABC

Step-5 Below AB, makes an acute angle

$\angle BAX = 60^0$

Step-6 AX, mark off five points

$A_1, A_2, A_3, A_4 and A_5$

Such that,

$AA_1 = A_1A_2 = A_2A_3 = A_3A_4 = A_4A_5$

Step-7 Join

A_5B

Step -8

Since we have to construct a triangle each of whose sides is three-five of the corresponding sides of

$\vartriangle ABC$

So, we take three parts out of five equal parts on AX from point

A_3

Draw

$B'C \parallel BC$

And meeting AC at C’

Thus,

$\triangle AB'C'$

Hence, the required triangle, each of whose side is $\frac{3}{5}$ of the corresponding sides of

$\triangle ABC$

23. Find the area of corresponding major sector of a circle with radius 7 cm and angle 120°.

Solution:

Given that,

$$\theta = 120^0$$
$$r = 7cm$$

Now,

Area of corresponding major sector

$$= \frac{\pi r^2 \left(360 - \theta\right)}{360}$$
$$= \frac{22}{7} \times 7 \times 7 \times \frac{360 - 120}{360}$$
$$= 154 \times \frac{240}{360}$$
$$= \frac{308}{3} cm^2$$

24. A copper rod of radius 1 cm and length 2 cm is drawn into a wire of length 18 m of uniform thickness. Find the thickness of the wire.

Solution:

Given that,

Diameter (D)= 1

Radius (R) = $\frac{1}{2} cm$

Height of rod (H) = Length of rod = 2 cm

Volume of rod,

$= \pi r^2 h$

$= \pi \times \left(\frac{1}{2}\right)^2 \times 2cm^3$

Cylindrical wire of length (h)18m = (1800cm)

$\therefore$ Volume of cylindrical rod = volume of cylindrical wire

Let r be the radius of cross-section of the wire.

$\therefore \pi \times \left(\frac{1}{2}\right)^2 \times 2 = \pi \times r^2 \times 1800$

$\Rightarrow r^2 = \frac{1}{900}$

$\Rightarrow r = \frac{1}{30} cm$

Thickness of the wire $= \frac{1}{30} \times 2 = \frac{1}{15} cm$

25. Neeraj and Dheeraj are friends. Find the probability of their birthdays when

(i) Birthdays are different.

(ii) Birthdays are same.

Solution:

Let A denote the even that Neeraj and Dheeraj have same birthday and B denote the even that Neeraj and Dheeraj have different birthday.

Neeraj and Dheeraj have the same birthday,

Then their birthday could be any one of the days from 365 days of the year.

Therefore, the number of cases in favourable of event A = 1

Required probability, $\text{P(A)} = \frac{1}{365}$

As, we know that,

$P(A)+P(B)=1$

$\therefore \frac{1}{365}+P(B)=1$

$P(B)=1-\frac{1}{365}$

$=\frac{365-1}{365}=\frac{364}{365}$

Probability that Neeraj and Dheeraj have the same day birthday $=\frac{1}{365}$

Probability that Neeraj and Dheeraj have different day birthday $=\frac{364}{365}$

PART - D

26. The cost of 5 apples and 3 oranges is Rs. 35 and the cost of 2 apples and 4 oranges is Rs. 28. Formulate the problem algebraically and Solutionve it graphically.

Solution:

Consider,

Let cost of 1 apple = x RS

And cost of 1 orange = y RS

According to the Question,

Cost of 5 apple = cost of 3 organge= 35 RS

$5x+3y=35$ --------(i)

Again, cost of 2 apple and 4 orange = 28 RS

$2x+4y=28$

$x+2y=14$ -----------(ii)

Multiplying 5 in equation (ii)

$5x+10y=70$

Subtracting equation (i)and (ii)

$5x+3y=35$
$5x+10y=70$
$-\quad -\qquad -$

$\boxed{-7y=35}$

$y=-5$

From equation (i)

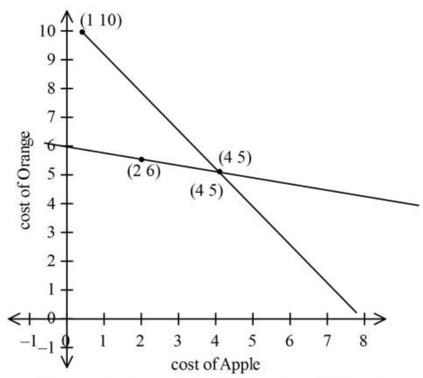
$5x+3y=35$
$5x=35-3y$
$x=\dfrac{35-3y}{5}$
$=\dfrac{35-15}{5}=\dfrac{20}{5}=4$
$and,$
$5x+3y=35$
$5x=35-3y$
$x=\dfrac{35-3y}{5}$
$x=\dfrac{35-3(10)}{5}=\dfrac{35-30}{5}=1$

X	4	1
Y	5	10

Again, from equation (ii)

$x+2y=14$
$x=14-2y$
$=14-2(5)$
$=14-10=4$
$and,$
$x+2y=14$
$x=14-2y$
$=14-2(6)$
$=14-12=2$

X	4	2
Y	5	6



Cost of Apple = 4

And cost of Orange = 5

27. The speed of a boat in still water is 18 km/h. It takes 1/ 2 an hour extra in going 12 km upstream instead of going the same distance downstream. Find the speed of the stream.

Solution:

Given that,

Speed of boat in still water = 18km/h

Distance = 12 km

Now,

Let speed of stream = x km/h

Let up stream = 18+x km/h

Downstream = 18-x km/h

Time is downstream – time is up stream = $\frac{1}{2}h$

$$\frac{12}{18-x}-\frac{12}{18+x}=\frac{1}{2}$$

Multiply by $(18-x)(18+x)$

$$(18-x)(18+x)\frac{12}{18-x}-(18-x)(18+x)\frac{12}{18+x}=\frac{1}{2}(18-x)(18+x)$$

$$216+12x-216+12x=\frac{1}{2}\left(18^2-x^2\right)$$

$$24x=\frac{1}{2}\left(324-x^2\right)$$

$$48x=324-x^2$$

$$x^2+48x-324$$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$=\frac{-48\pm\sqrt{48^2-4\times1\times(-324)}}{2\times1}$$

$$=\frac{-48\pm\sqrt{2304+1296}}{2}$$

$$=\frac{-48\pm\sqrt{3600}}{2}$$

$$=\frac{-48\pm60}{2}$$

Then,

$$x=\frac{-48+60}{2}$$

$$=\frac{12}{2}=6$$

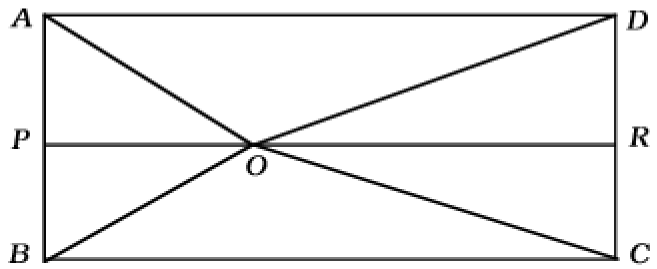
$$and,$$

$$x=\frac{-48-60}{2}$$

$$=\frac{-108}{2}=-54$$

Hence, the speed of stream = 6km/h.

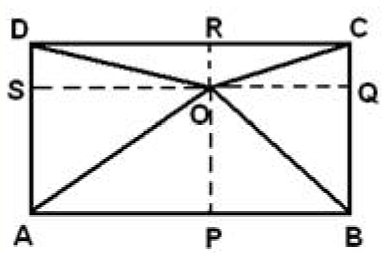
28. O is any point inside rectangle ABCD. Prove that $OB^2 + OD^2 = OA^2 + OC^2$



Solution:

Consider,

The point O construct perpendiculars OP, OQ, OR and OS to the sides of the rectangle



$RHS = OA^2 + OC^2$

$= (AS^2 + OS^2) + (OQ^2 + QC^2)$ (considering right triangle ASO and COQ)

But AS = BQ and OQ = BP

Therefore,

$(AS^2 + OS^2) + (OQ^2 + QC^2)$

$= (BQ^2 + OS^2) + (OQ^2 + DS^2)$

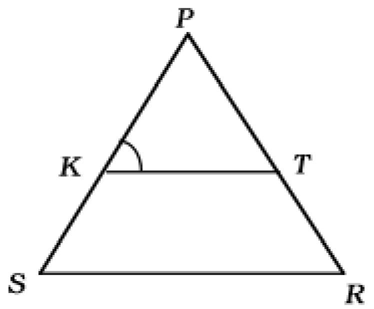
$= (BQ^2 + OQ^2) + (OS^2 + DS^2)$

$= OB^2 + OD^2$ (considering right triangle BOQ and ODS)

= LHS

OR,

In the given figure, $\frac{\text{PK}}{\text{KS}} = \frac{\text{PT}}{\text{TR}}$ and $\angle\text{PKT} = \angle\text{PRS}$. Prove that ΔPSR is an isosceles triangle.



Solution:

Given that,

$$\frac{PK}{KS} = \frac{PT}{TR}$$
$$\angle PKT = \angle PRS$$

Now,

In $\triangle PSR$

$$\frac{PK}{KS} = \frac{PT}{TR} \quad \text{(given)}$$
$$\text{KT} \parallel \text{SR}$$
$$\angle\text{PKT} = \angle\text{PSR}\text{--------(i)}$$
$$\angle\text{PKT} = \angle\text{PRS} \quad \text{(given)------(ii)}$$

From equation (i) and (ii)

$$\angle\text{PSR} = \angle\text{PRS}$$

So, the triangle PSR is a isosceles triangle.

29. Find the value of k if the points A (2, 3), B (4, k) and C (6, – 3) are collinear.

Solution:

Given that,

$A(2,3), B(4,k)$ and $C(6,-3)$ point are collinear

Then,

Area of triangle,

$$= \frac{1}{2}\left[x_1(y_1 - y_2) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right]$$

$$0 = \frac{1}{2}\left[2(k+3) + 4(-3-3) + 6(3-k)\right]$$

$$= 2k + 6 - 24 + 18 - 6k$$

$$= -4k - 24 + 24$$

$$= -4k$$

$$k = \frac{0}{4} = 0$$

Hence, the value of k = 0.

30. In the following distribution calculate mean $\overline{x}$ from assumed mean:

Class-interval	**10-25**	**25-40**	**40-55**	**55-70**	**70-85**	**85-100**
Frequency	**2**	**3**	**4**	**5**	**6**	**7**

Solution:

Class	Frequency f_i	Mid value x_i	$u_i = \frac{(x_i - A)}{h}$	$(f_i \times u_i)$
10-25	2	17.5	-2	-4
25-40	3	32.5	-1	-3
40-55	7	47.5	0	0
55-70	5	62.5	+1	5
70-85	6	77.5	+2	12
85-100	7	92.5	+3	21
	$\sum f_i = 30$			$\sum(f_i \times u_i) = 31$

Thus,

$A = 30, h = 15, \sum f_i = 27$ and $\sum \left(f_i \times u_i \right) = 31$

$$mean(\overline{x}) = A + \left\{ h \times \frac{\sum \left(f_i \times u_i \right)}{\sum f_i} \right\}$$

$$= 45.5 + \frac{31}{30} \times 15$$

$$= 45.5 + 15.5$$

$$= 63$$

OR,

Find the mode of the following distribution:

Class-interval	0-20	20-40	40-60	60-80	80-100	100-120
Frequency	**10**	**35**	**52**	**61**	**38**	**20**

Solution:

As the class 40-60 has maximum frequency, so it is the modal class.

$x_k = 60, h = 20, f_k = 61, f_{k-1} = 52$ and $f_{k+1} = 38$

$$M_o = x_k + \left\{ h \times \frac{(f_k - f_{k-1})}{(2f_k - f_{k-1} - f_{k+1})} \right\}$$

$$= 60 + 20 \times \frac{61 - 52}{2 \times 61 - 52 - 38}$$

$$= 60 + 20 \times \frac{9}{122 - 90}$$

$$= 60 + 20 \times \frac{9}{32}$$

$$= 60 + \frac{45}{8}$$

$$= 60 + 5.625$$

$$= 65.625$$